Sri Lanka Institute of Information Technology

Sky Survivor

2D Interactive Game - Assignment 02

SE4031 - Games Development

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Game Concept

This project is a 2D interactive game developed using Unity. The player controls a flying character that must navigate through obstacles while avoiding enemies and collecting health packs. The objective is to survive as long as possible and achieve a high score.

The game includes dynamic gameplay elements such as increasing speed, real-time health management, enemy interactions, and shooting mechanics. Visual and audio feedback systems enhance the overall player experience.

Implementation of Core Features

Health System

The game includes a health system displayed using a UI health bar. The player loses health when colliding with obstacles or enemies. Health is limited to a maximum value and decreases gradually until reaching zero, which triggers the game over state.

Score System

The score increases when the player successfully passes obstacles and destroys enemies. The score is displayed on the screen and resets when the game restarts.

Health Packs

Health packs appear during gameplay and restore player health upon collection. They are automatically destroyed after being collected or when they move off-screen.

Projectile System

The player can shoot projectiles using the left mouse button. Projectiles destroy enemies upon collision and are also removed after a short duration.

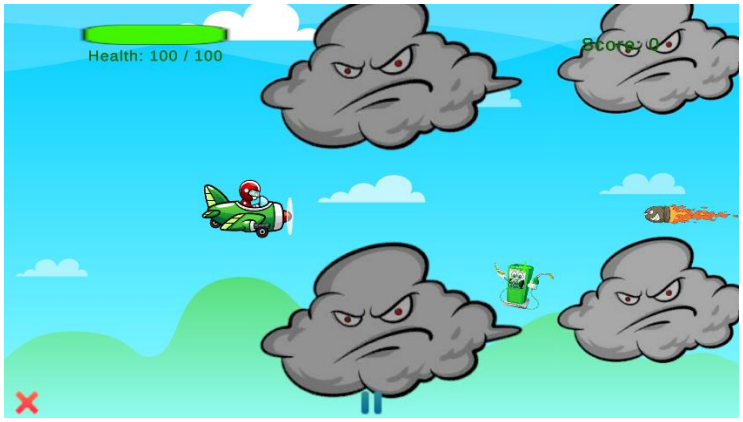
Enemy System

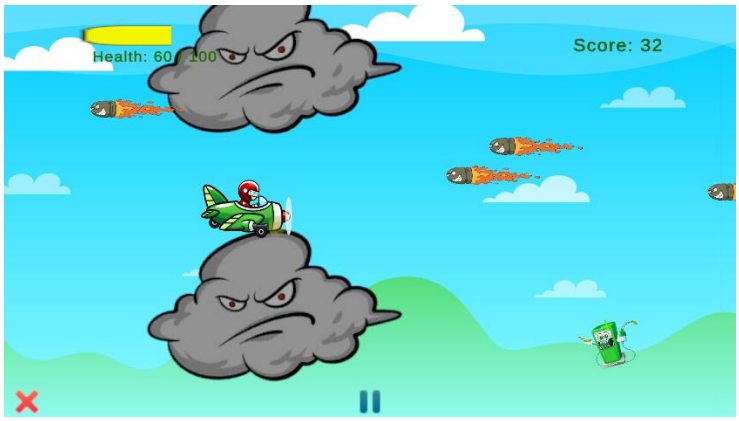
Enemies move toward the player and cause damage on collision. They can be destroyed using projectiles, which also increases the score. Enemies are removed after passing the player.

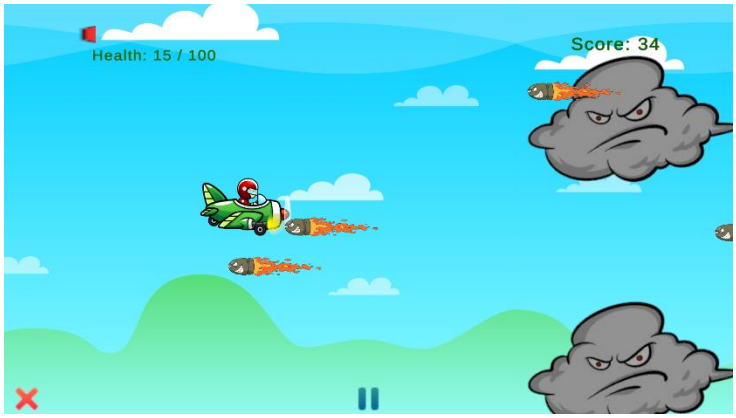
Speed Increase

The game speed gradually increases over time to enhance difficulty. This creates a progressively challenging gameplay experience.

Screenshots






Health: 15 / 100
Score: 34

Health: 0 / 100
Score: 34
GAME OVER
Press R to Restart

Controls

- Left Mouse Click - Shoot projectile
- ESC - Pause / Resume game
- Q - Quit game
- R - Restart game (after Game Over)

Creative Features

The game includes several creative enhancements to improve user experience:

- Red screen flash effect when taking damage
- Camera shake effect on collisions
- Explosion (blast) effect when enemies are destroyed
- Smooth speed increase over time
- Animated background and themed visual assets

Credits

Game developed using Unity Engine.

Assets used:

- Free sprite assets (OpenGameArt / Kenney / etc.)
- Audio effects from free online sources

All assets are used for educational purposes.

Gameplay Video Link

https://drive.google.com/file/d/1MsBeT2PPCfGTL84Z48D1T4DWazk14zZC/view?usp=sharing